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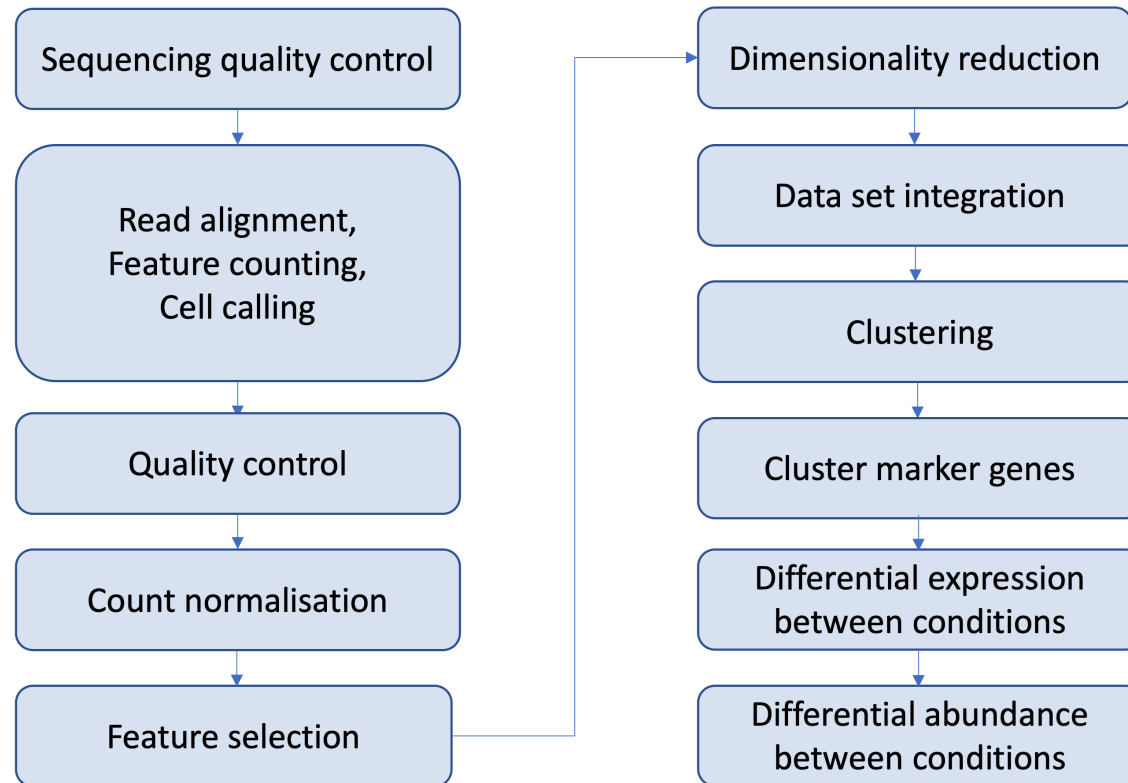
Introduction to single-cell RNA-seq analysis - Normalisation and Feature Selection

February 2026

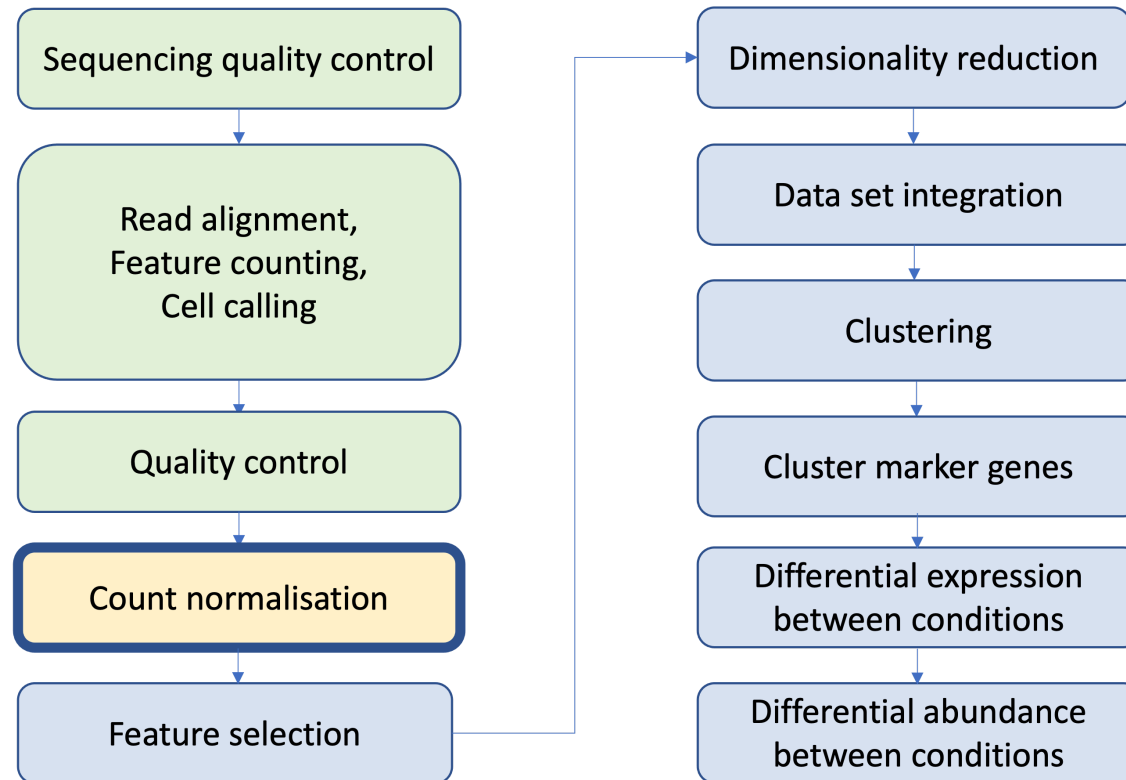
Outline

- Motivation
- Biases
 - Depth bias
 - Composition bias
 - Mean-variance correlation
- Normalisation strategies
- Feature Selection

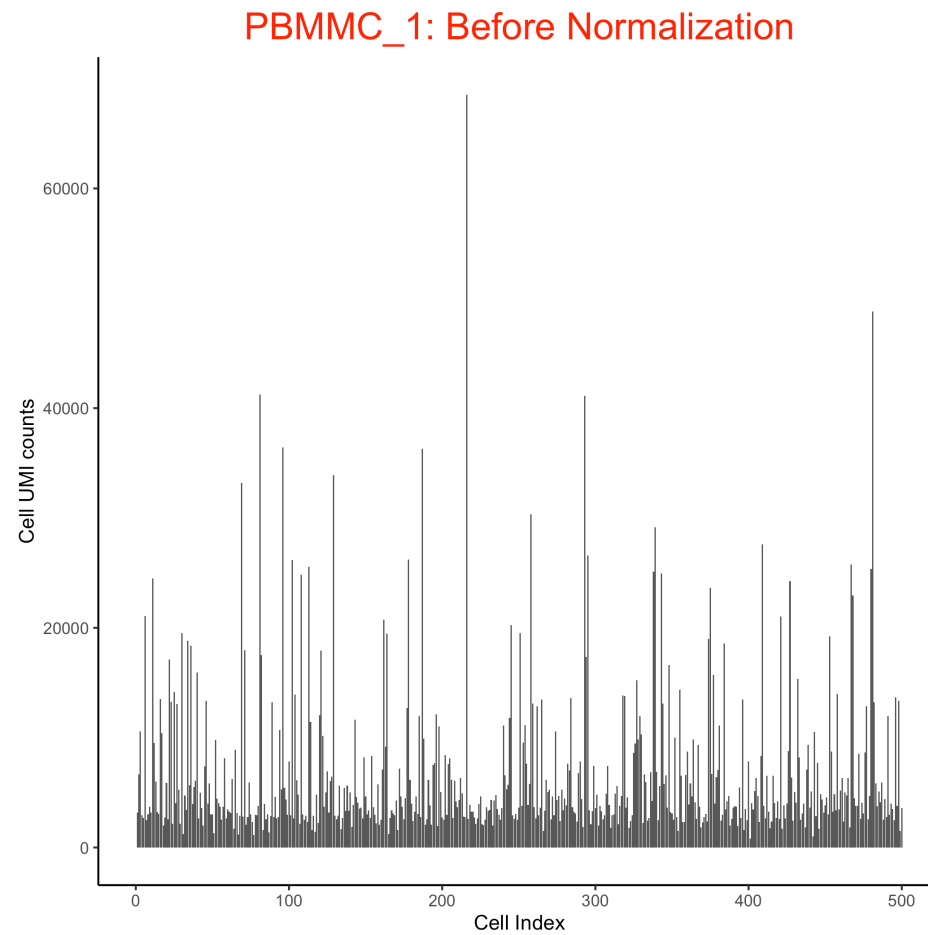
Workflow



Workflow



Raw UMI counts distribution



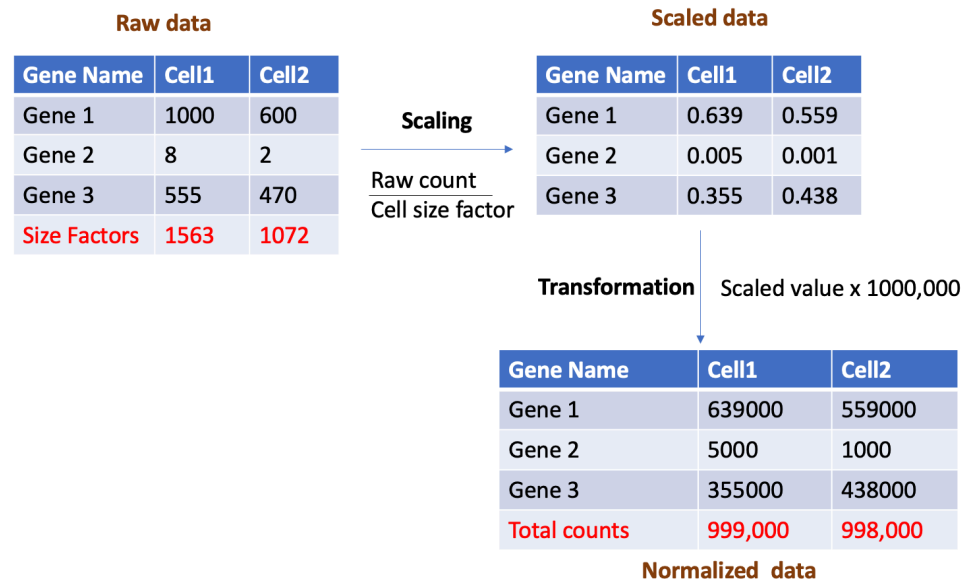
Why do UMI counts differ among the cells?

- We derive biological insights downstream by comparing cells against each other.
- But the UMI count differences makes it harder to compare cells.
- Why do total transcript molecules (UMI counts) detected between cells differ?
 - Biological:
 - Cell subtype differences - size and transcriptional activity, variation in gene expression
 - Technical: scRNA data is inherently noisy
 - Low mRNA content per cell
 - cell-to-cell differences in mRNA capture efficiency
 - Variable sequencing depth
 - PCR amplification efficiency

Normalization reduces technical differences so that differences between cells are not technical but biological, allowing meaningful comparison of expression profiles between cells.

Depth bias

Depth bias: Read differences between cells



Simple library size normalization accounts for the depth bias

Composition bias

- Few genes contribute to most read counts
- In this example, the total read counts are the same across the cells
- Gene 1 contributes 80% of reads in cell2, leaving other genes with fewer read counts.

Gene Name	Cell1	Cell2
Gene 1	333	800
Gene 2	333	100
Gene 3	334	100
Total read counts	1000	1000

- Library size normalization can not correct composition bias.

Mean-variance correlation

Mean and variance of raw counts for genes are correlated

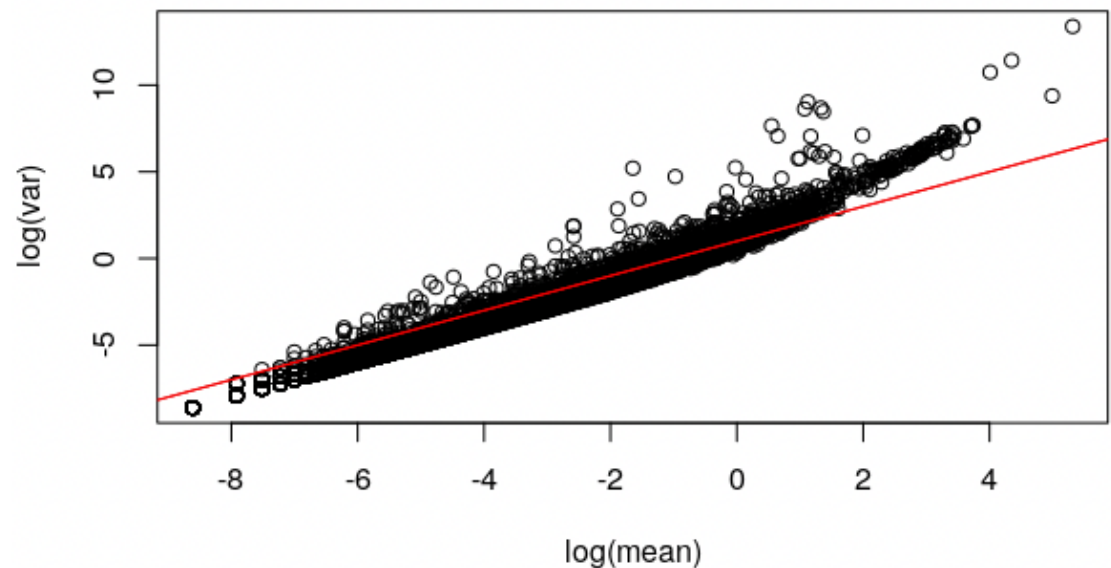
More highly expressed genes tend to look more variable because larger numbers result in higher variance

A gene expressed at a low level tends to have a low variance across cells:

$$\text{var}(c(2,4,2,4,2,4,2,4)) = 1.14$$

A gene with the same proportional differences between cells, but expressed at a higher level will have higher variance:

$$\text{var}(c(20,40,20,40,20,40,20,40)) = 114.29$$



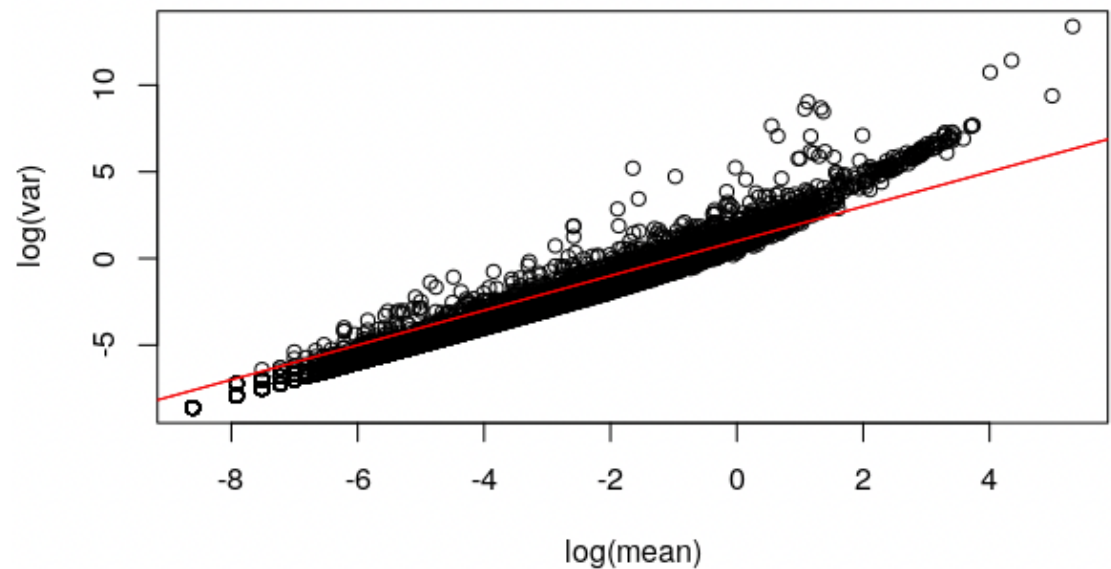
Mean-variance correlation

If we take the logs of the expression values, the variances are the same for both genes:

$$\text{var}(\log(c(2,4,2,4,2,4,2,4))) = 0.14$$

$$\text{var}(\log(c(20,40,20,40,20,40,20,40))) = 0.14$$

This “variable stabilising transformation” helps to remove the correlation between mean and variance



General principle behind normalisation

Normalization has two steps

1. Scaling

- Calculate size factors or normalization factors that represents the relative depth bias in each cell
- Scale the counts for each gene in each cell by dividing the raw counts with cell specific size factor

2. Transformation: Transform the data after scaling

- Per million (e.g. CPM)
- $\log_2$ (e.g. Deconvolution)
- Pearson residuals (eg. SCTransform)

Bulk RNAseq methods are not suitable for scRNAseq data

CPM: convert raw counts to counts-per-million (CPM)

- for each cell
- by dividing counts by the library size then multiplying by 1.000.000.
- does not address compositional bias caused by highly expressed genes that are also differentially expressed between cells.

DESeq's size factor

- For each gene, compute geometric mean across cells
- For each cell
 - compute for each gene the ratio of its expression to its geometric mean,
 - derive the cell's size factor as the median ratio across genes.
- Not suitable for sparse scRNA-seq data as the geometric mean is computed on non-zero values only.

SCTransform

“This procedure omits the need for heuristic steps including pseudocount addition or log-transformation and improves common downstream analytical tasks such as variable gene selection, dimensional reduction, and differential expression. We named this method sctransform.”

Steps:

1. Estimate size factors using a regularized negative binomial regression model
2. Scale the counts by dividing the raw counts with the estimated size factors
3. Apply a variance stabilizing transformation to the scaled counts by calculating Pearson residuals from the negative binomial regression model

SCTransform

Pearson residual = (Observed value – Expected value) / $\sqrt{\text{Expected variance}}$

If a gene has exactly the expression level we'd expect based on the model → residual = 0

If a gene is expressed higher than expected → positive residual

If a gene is expressed lower than expected → negative residual

These residuals become our normalised values and they're useful for several reasons:

- Standardized scale: All genes end up on the same scale regardless of their expression level
- Variance stabilization: Unlike raw counts, these residuals have roughly the same variance across all expression levels. The transformation ensures that genes with different expression levels can be compared fairly – the variance no longer depends on the mean expression level.
- Interpretation: Positive residuals mean higher-than-expected expression; negative means lower-than-expected. A residual of +2 means “this gene is expressed about 2 standard deviations higher than expected”.

SCTransform

Really Useful Website:

<https://biostatsquid.com/sctransform-simple-explanation/>

Inside SCTransform there is a 'selection' of the the most variable genes.

By default these are used for downstream dimensionality reduction and clustering (although in most cases you can change this)

- Which genes should we use for downstream analysis?

Select genes which capture biologically-meaningful variation, while reducing the number of genes which only contribute to technical noise

Recap

- We get different total counts for each cell due to technical factors (depth bias)
- A simplistic library size normalisation (e.g. CPM) removes a large part of this bias
- However, composition bias causes spurious differences between cells
- Early methods developed for bulk RNA-seq are not appropriate for sparse scRNA- seq data.
- The SCTransform method uses a regularized negative binomial regression model to estimate size factors, and applies a variance stabilizing transformation to the scaled counts.
- This accounts for both sequencing depth (library size) and gene-specific effects simultaneously.